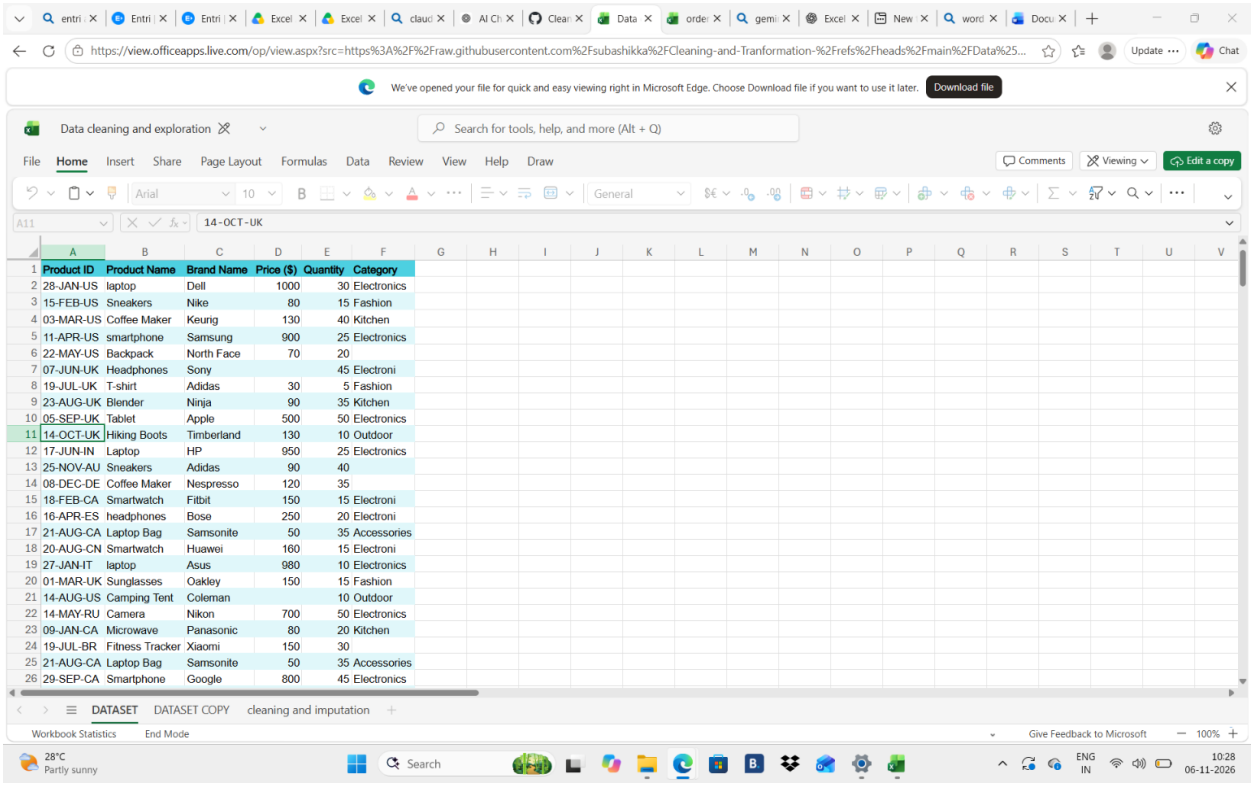
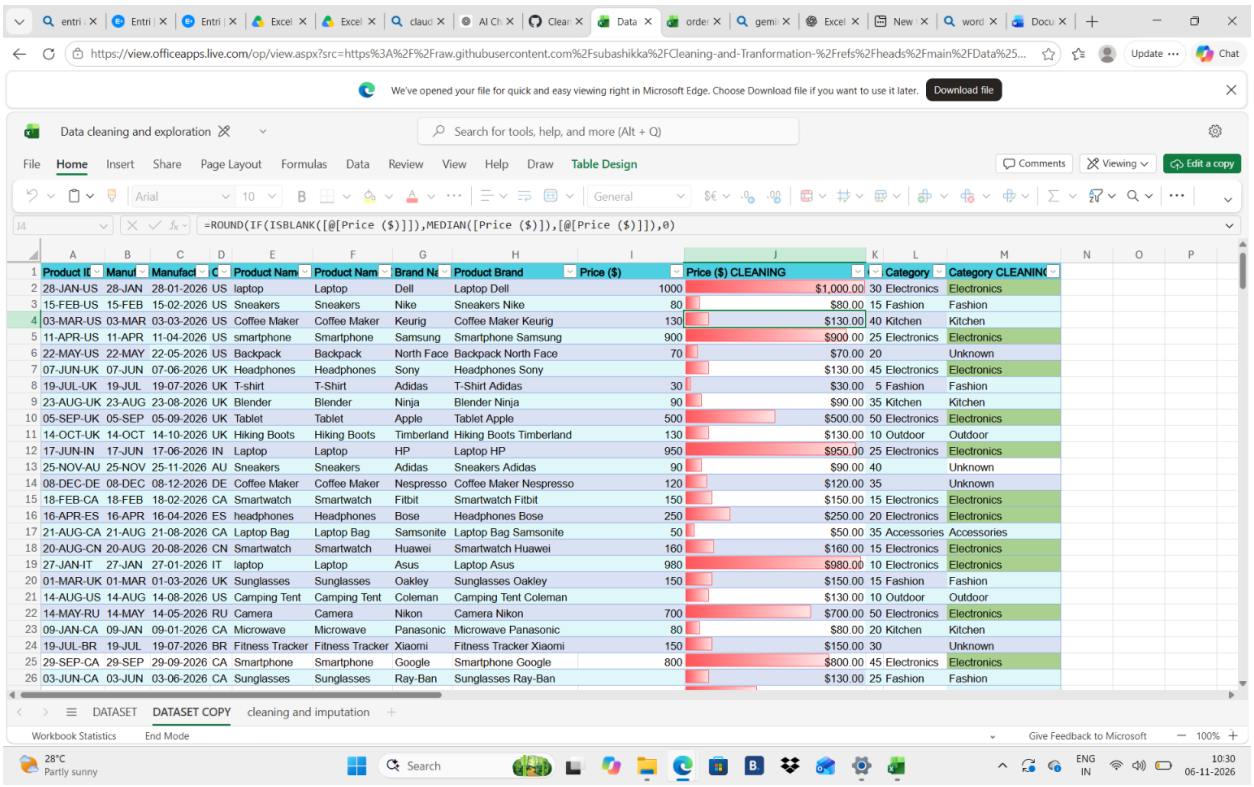
Dataset Description

Tasks:1

1. Handling Missing Values:
   - Check for missing values in the 'Price' column. How would you handle products with missing price information?

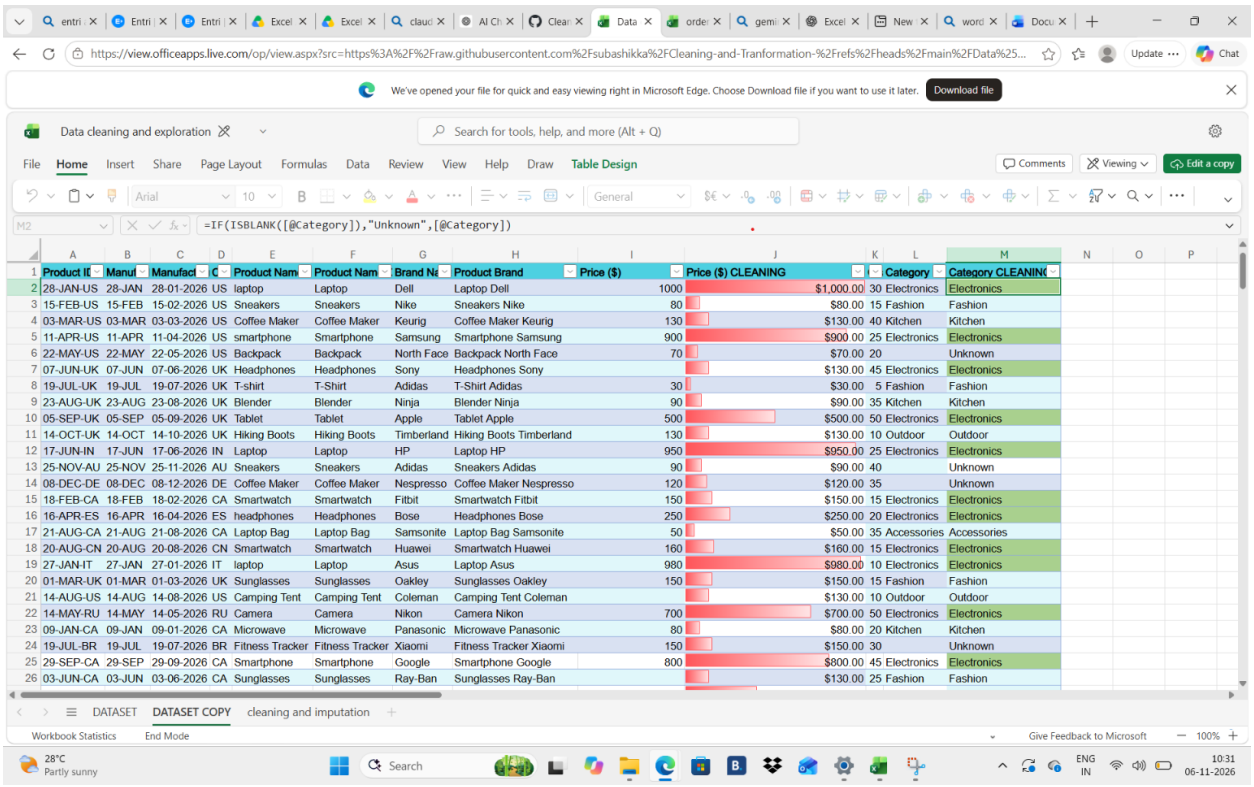


	A	B	C	D	E	F
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
2	28-JAN-US	laptop	Dell	1000	30	Electronics
3	15-FEB-US	Sneakers	Nike	80	15	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
5	11-APR-US	smartphone	Samsung	900	25	Electronics
6	22-MAY-US	Backpack	North Face	70	20	
7	07-JUN-UK	Headphones	Sony		45	Electroni
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen
10	05-SEP-UK	Tablet	Apple	500	50	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
12	17-JUN-IN	Laptop	HP	950	25	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	40	
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16	16-APR-ES	headphones	Bose	250	20	Electroni
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
19	27-JAN-IT	laptop	Asus	980	10	Electronics
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor
22	14-MAY-RU	Camera	Nikon	700	50	Electronics
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
26	29-SEP-CA	Smartphone	Google	800	45	Electronics



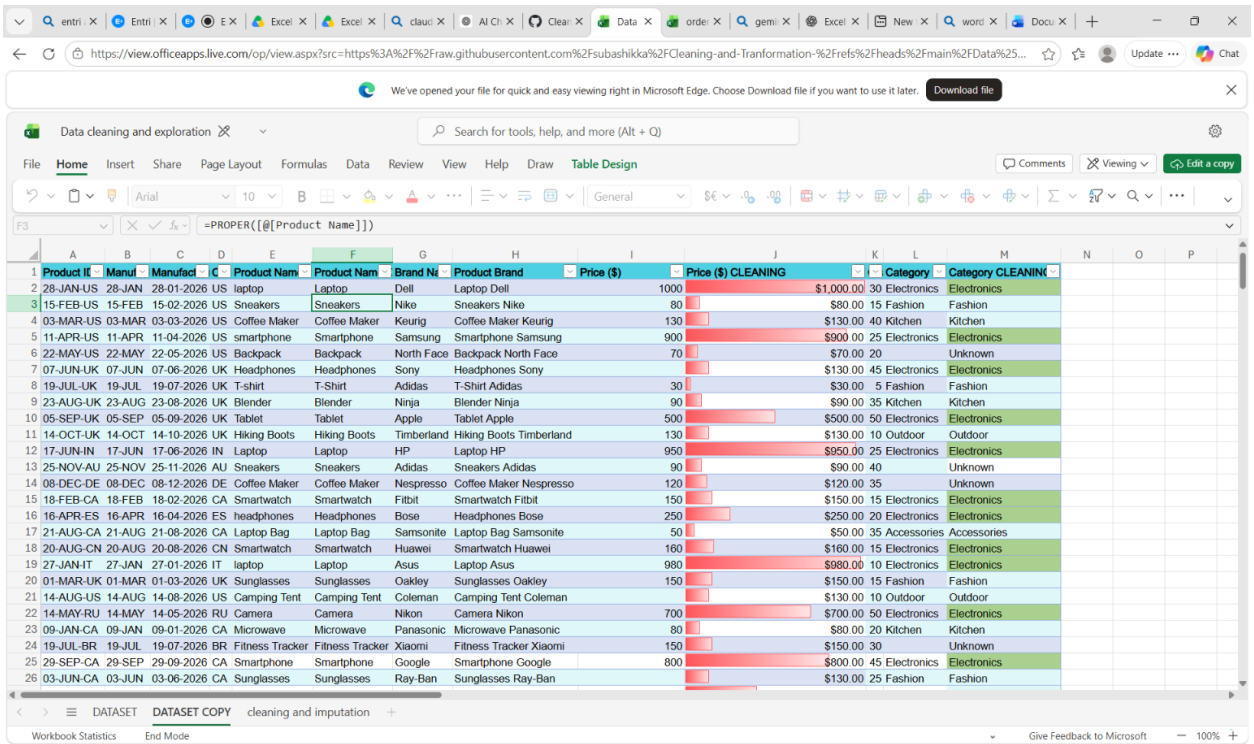
Product ID	Manuf	Manufact	C	Product Nam	Product Nam	Brand Na	Product Brand	Price ($)	Price ($) CLEANING		Category	Category CLEANIN(
28-JAN-US	28-JAN	28-01-2026	US	laptop	Laptop	Dell	Laptop Dell	1000	$1,000.00	30	Electronics	Electronics
15-FEB-US	15-FEB	15-02-2026	US	Sneakers	Sneakers	Nike	Sneakers Nike	80	$80.00	15	Fashion	Fashion
03-MAR-US	03-MAR	03-03-2026	US	Coffee Maker	Coffee Maker	Keurig	Coffee Maker Keurig	130	$130.00	40	Kitchen	Kitchen
11-APR-US	11-APR	11-04-2026	US	smartphone	Smartphone	Samsung	Smartphone Samsung	900	$900.00	25	Electronics	Electronics
22-MAY-US	22-MAY	22-05-2026	US	Backpack	Backpack	North Face	Backpack North Face	70	$70.00	20		Unknown
07-JUN-UK	07-JUN	07-06-2026	UK	Headphones	Headphones	Sony	Headphones Sony		$130.00	45	Electronics	Electronics
19-JUL-UK	19-JUL	19-07-2026	UK	T-shirt	T-Shirt	Adidas	T-Shirt Adidas	30	$30.00	5	Fashion	Fashion
23-AUG-UK	23-AUG	23-08-2026	UK	Blender	Blender	Ninja	Blender Ninja	90	$90.00	35	Kitchen	Kitchen
05-SEP-UK	05-SEP	05-09-2026	UK	Tablet	Tablet	Apple	Tablet Apple	500	$500.00	50	Electronics	Electronics
14-OCT-UK	14-OCT	14-10-2026	UK	Hiking Boots	Hiking Boots	Timberland	Hiking Boots Timberland	130	$130.00	10	Outdoor	Outdoor
17-JUN-IN	17-JUN	17-06-2026	IN	Laptop	Laptop	HP	Laptop HP	950	$950.00	25	Electronics	Electronics
25-NOV-AU	25-NOV	25-11-2026	AU	Sneakers	Sneakers	Adidas	Sneakers Adidas	90	$90.00	40		Unknown
08-DEC-DE	08-DEC	08-12-2026	DE	Coffee Maker	Coffee Maker	Nespresso	Coffee Maker Nespresso	120	$120.00	35		Unknown
18-FEB-CA	18-FEB	18-02-2026	CA	Smartwatch	Smartwatch	Fitbit	Smartwatch Fitbit	150	$150.00	15	Electronics	Electronics
16-APR-ES	16-APR	16-04-2026	ES	headphones	Headphones	Bose	Headphones Bose	250	$250.00	20	Electronics	Electronics
21-AUG-CA	21-AUG	21-08-2026	CA	Laptop Bag	Laptop Bag	Samsonite	Laptop Bag Samsonite	50	$50.00	35	Accessories	Accessories
20-AUG-CN	20-AUG	20-08-2026	CN	Smartwatch	Smartwatch	Huawei	Smartwatch Huawei	160	$160.00	15	Electronics	Electronics
27-JAN-IT	27-JAN	27-01-2026	IT	laptop	Laptop	Asus	Laptop Asus	980	$980.00	10	Electronics	Electronics
01-MAR-UK	01-MAR	01-03-2026	UK	Sunglasses	Sunglasses	Oakley	Sunglasses Oakley	150	$150.00	15	Fashion	Fashion
14-AUG-US	14-AUG	14-08-2026	US	Camping Tent	Camping Tent	Coleman	Camping Tent Coleman		$130.00	10	Outdoor	Outdoor
14-MAY-RU	14-MAY	14-05-2026	RU	Camera	Camera	Nikon	Camera Nikon	700	$700.00	50	Electronics	Electronics
09-JAN-CA	09-JAN	09-01-2026	CA	Microwave	Microwave	Panasonic	Microwave Panasonic	80	$80.00	20	Kitchen	Kitchen
19-JUL-BR	19-JUL	19-07-2026	BR	Fitness Tracker	Fitness Tracker	Xiaomi	Fitness Tracker Xiaomi	150	$150.00	30		Unknown
29-SEP-CA	29-SEP	29-09-2026	CA	Smartphone	Smartphone	Google	Smartphone Google	800	$800.00	45	Electronics	Electronics
03-JUN-CA	03-JUN	03-06-2026	CA	Sunglasses	Sunglasses	Ray-Ban	Sunglasses Ray-Ban		$130.00	25	Fashion	Fashion

- If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.



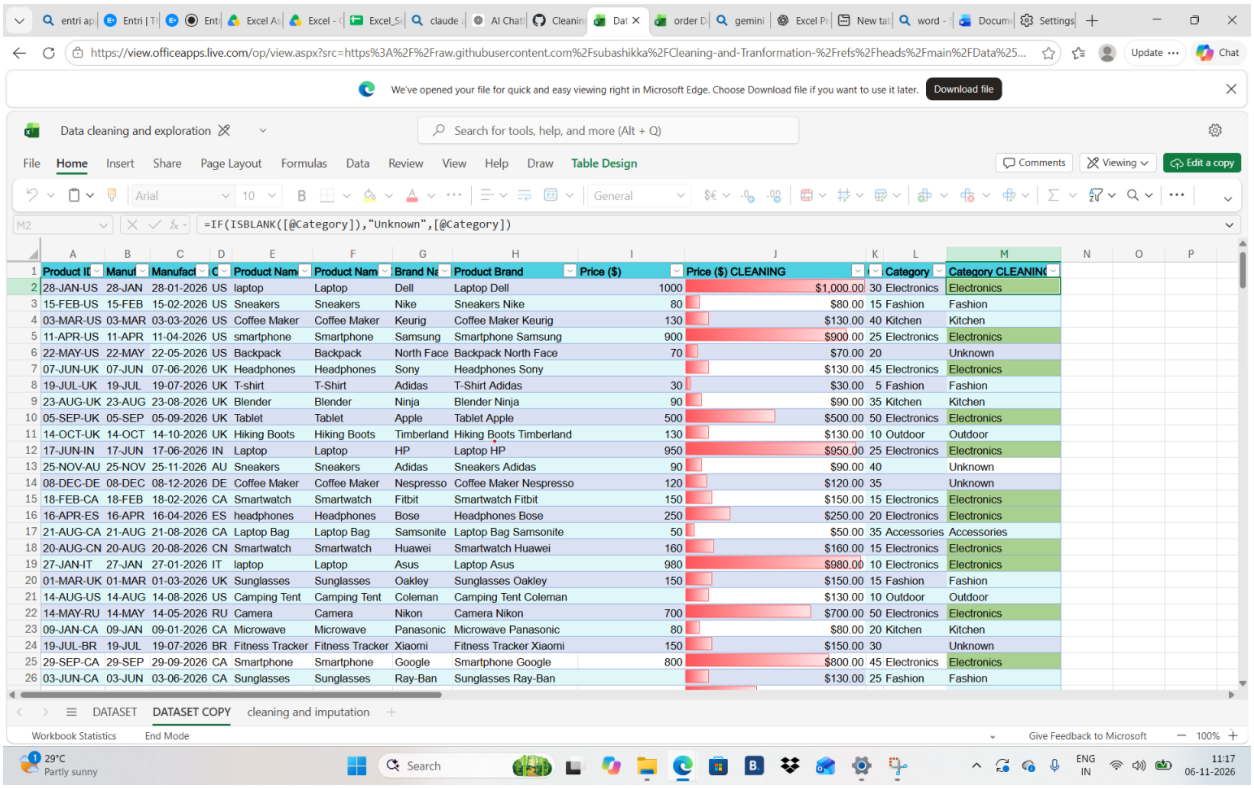
2. Correcting Inconsistent Data:

● Identify any inconsistent text formats present in the "Product Name" column.



● Identify any typos present in the "Category" column.

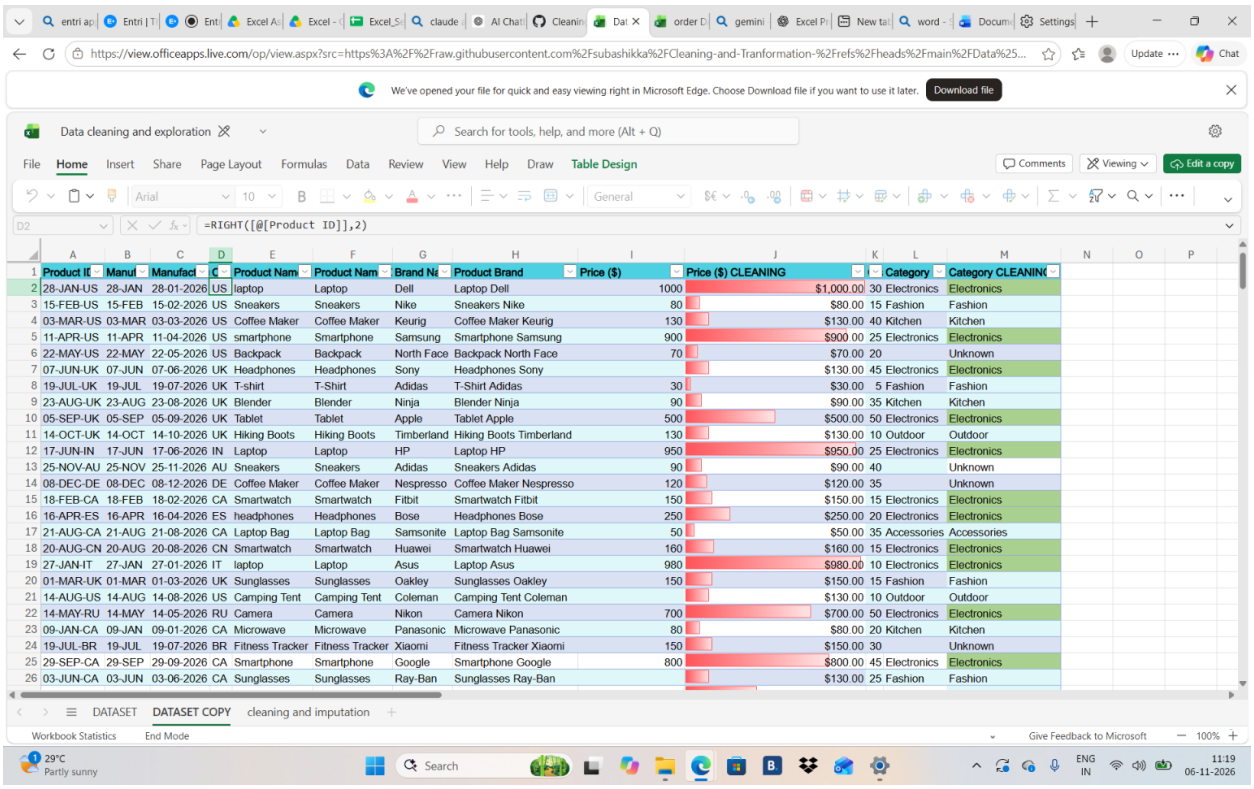
● Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.

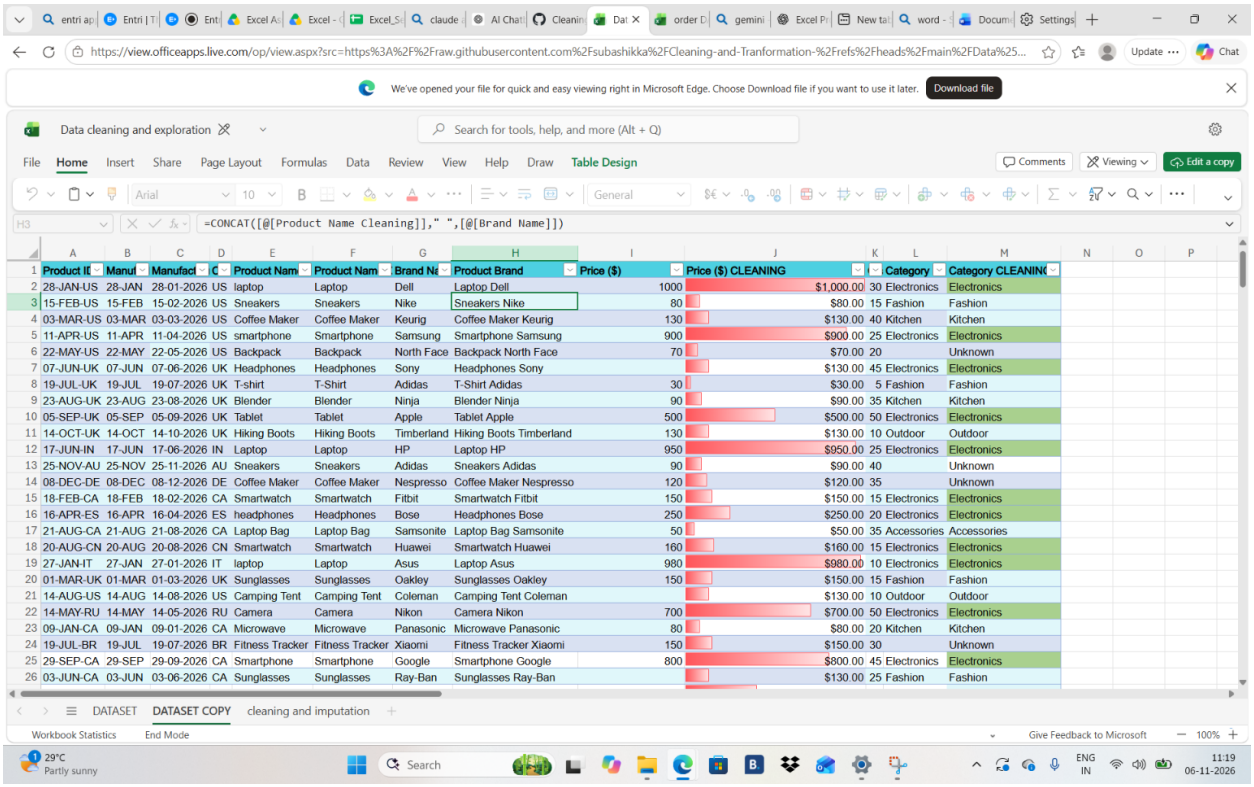
3. 

Product ID	Manuf	Manufact	C	Product Nam	Product Nam	Brand Na	Product Brand	Price ($)	Price ($) CLEANING		Category	Category CLEANING
28-JAN-US	28-JAN	28-01-2026	US	laptop	Laptop	Dell	Laptop Dell	1000	$1,000.00	30	Electronics	Electronics
15-FEB-US	15-FEB	15-02-2026	US	Sneakers	Sneakers	Nike	Sneakers Nike	80	$80.00	15	Fashion	Fashion
03-MAR-US	03-MAR	03-03-2026	US	Coffee Maker	Coffee Maker	Keurig	Coffee Maker Keurig	130	$130.00	40	Kitchen	Kitchen
11-APR-US	11-APR	11-04-2026	US	smartphone	Smartphone	Samsung	Smartphone Samsung	900	$900.00	25	Electronics	Electronics
22-MAY-US	22-MAY	22-05-2026	US	Backpack	Backpack	North Face	Backpack North Face	70	$70.00	20		Unknown
07-JUN-UK	07-JUN	07-06-2026	UK	Headphones	Headphones	Sony	Headphones Sony		$130.00	45	Electronics	Electronics
19-JUL-UK	19-JUL	19-07-2026	UK	T-shirt	T-Shirt	Adidas	T-Shirt Adidas	30	$30.00	5	Fashion	Fashion
23-AUG-UK	23-AUG	23-08-2026	UK	Blender	Blender	Ninja	Blender Ninja	90	$90.00	35	Kitchen	Kitchen
05-SEP-UK	05-SEP	05-09-2026	UK	Tablet	Tablet	Apple	Tablet Apple	500	$500.00	50	Electronics	Electronics
14-OCT-UK	14-OCT	14-10-2026	UK	Hiking Boots	Hiking Boots	Timberland	Hiking Boots Timberland	130	$130.00	10	Outdoor	Outdoor
17-JUN-IN	17-JUN	17-06-2026	IN	Laptop	Laptop	HP	Laptop HP	950	$950.00	25	Electronics	Electronics
25-NOV-AU	25-NOV	25-11-2026	AU	Sneakers	Sneakers	Adidas	Sneakers Adidas	90	$90.00	40		Unknown
08-DEC-DE	08-DEC	08-12-2026	DE	Coffee Maker	Coffee Maker	Nespresso	Coffee Maker Nespresso	120	$120.00	35		Unknown
18-FEB-CA	18-FEB	18-02-2026	CA	Smartwatch	Smartwatch	Fitbit	Smartwatch Fitbit	150	$150.00	15	Electronics	Electronics
16-APR-ES	16-APR	16-04-2026	ES	headphones	Headphones	Bose	Headphones Bose	250	$250.00	20	Electronics	Electronics
21-AUG-CA	21-AUG	21-08-2026	CA	Laptop Bag	Laptop Bag	Samsonite	Laptop Bag Samsonite	50	$50.00	35	Accessories	Accessories
20-AUG-CN	20-AUG	20-08-2026	CN	Smartwatch	Smartwatch	Huawei	Smartwatch Huawei	160	$160.00	15	Electronics	Electronics
27-JAN-IT	27-JAN	27-01-2026	IT	laptop	Laptop	Asus	Laptop Asus	980	$980.00	10	Electronics	Electronics
01-MAR-UK	01-MAR	01-03-2026	UK	Sunglasses	Sunglasses	Oakley	Sunglasses Oakley	150	$150.00	15	Fashion	Fashion
14-AUG-US	14-AUG	14-08-2026	US	Camping Tent	Camping Tent	Coleman	Camping Tent Coleman		$130.00	10	Outdoor	Outdoor
14-MAY-RU	14-MAY	14-05-2026	RU	Camera	Camera	Nikon	Camera Nikon	700	$700.00	50	Electronics	Electronics
09-JAN-CA	09-JAN	09-01-2026	CA	Microwave	Microwave	Panasonic	Microwave Panasonic	80	$80.00	20	Kitchen	Kitchen
19-JUL-BR	19-JUL	19-07-2026	BR	Fitness Tracker	Fitness Tracker	Xiaomi	Fitness Tracker Xiaomi	150	$150.00	30		Unknown
29-SEP-CA	29-SEP	29-09-2026	CA	Smartphone	Smartphone	Google	Smartphone Google	800	$800.00	45	Electronics	Electronics
03-JUN-CA	03-JUN	03-06-2026	CA	Sunglasses	Sunglasses	Ray-Ban	Sunglasses Ray-Ban		$130.00	25	Fashion	Fashion

4. Removing Duplicates:
   ● Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

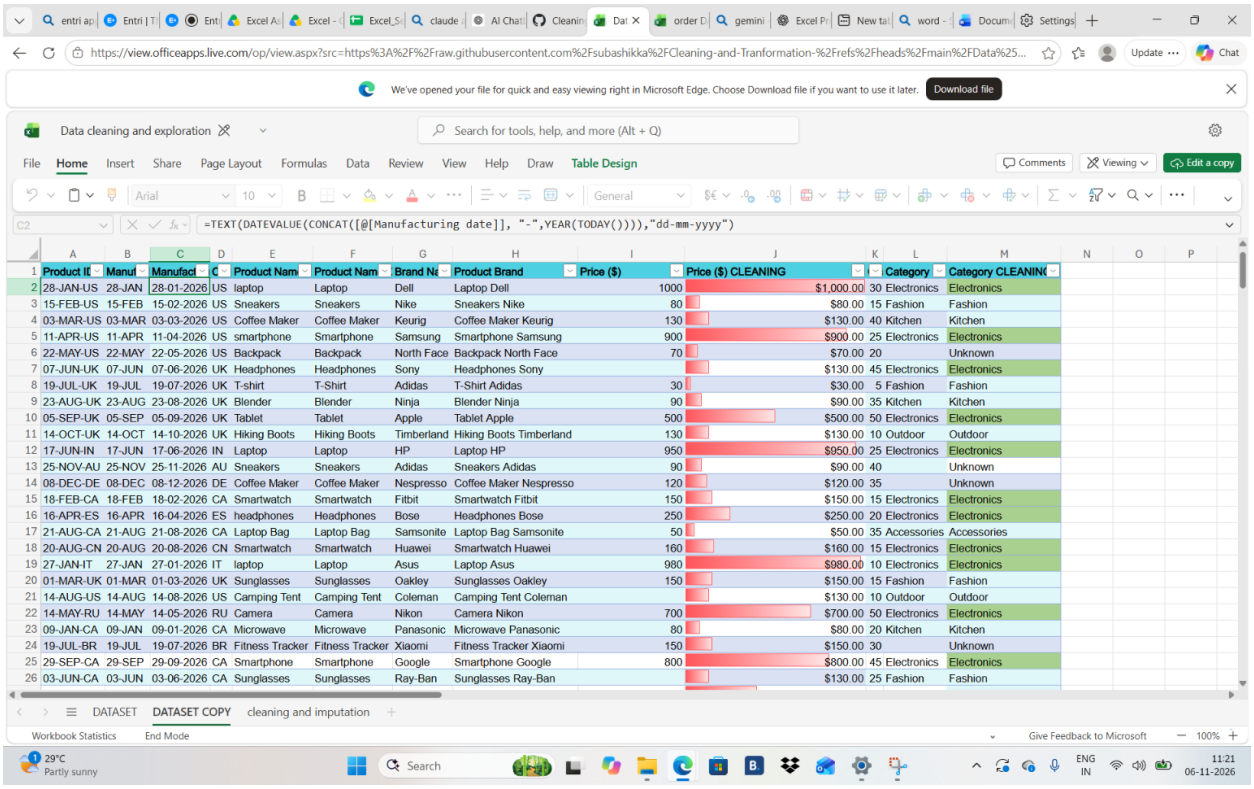
5. Splitting and Merging Data:
   ● Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.
   ● Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".





5. Number Formatting:
   - Format the data type of the "Price" column to a currency format.

● Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

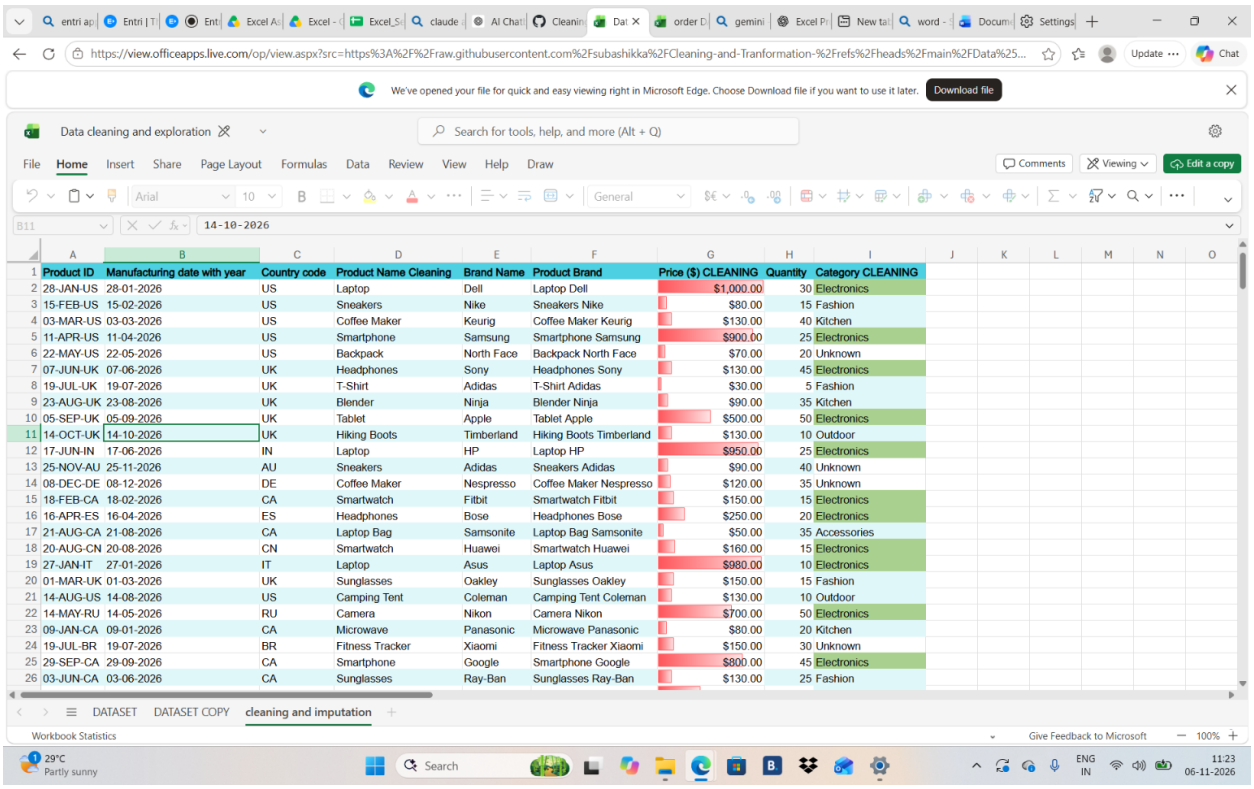


Product ID	Manuf	Manufac	C	Product Nam	Product Nam	Brand Na	Product Brand	Price ($)	Price ($) CLEANING		Category	Category CLEANING
28-JAN-US	28-JAN	28-01-2026	US	laptop	Laptop	Dell	Laptop Dell	1000	$1,000.00	30	Electronics	Electronics
15-FEB-US	15-FEB	15-02-2026	US	Sneakers	Sneakers	Nike	Sneakers Nike	80	$80.00	15	Fashion	Fashion
03-MAR-US	03-MAR	03-03-2026	US	Coffee Maker	Coffee Maker	Keurig	Coffee Maker Keurig	130	$130.00	40	Kitchen	Kitchen
11-APR-US	11-APR	11-04-2026	US	smartphone	Smartphone	Samsung	Smartphone Samsung	900	$900.00	25	Electronics	Electronics
22-MAY-US	22-MAY	22-05-2026	US	Backpack	Backpack	North Face	Backpack North Face	70	$70.00	20		Unknown
07-JUN-UK	07-JUN	07-06-2026	UK	Headphones	Headphones	Sony	Headphones Sony		$130.00	45	Electronics	Electronics
19-JUL-UK	19-JUL	19-07-2026	UK	T-shirt	T-Shirt	Adidas	T-Shirt Adidas	30	$30.00	5	Fashion	Fashion
23-AUG-UK	23-AUG	23-08-2026	UK	Blender	Blender	Ninja	Blender Ninja	90	$90.00	35	Kitchen	Kitchen
05-SEP-UK	05-SEP	05-09-2026	UK	Tablet	Tablet	Apple	Tablet Apple	500	$500.00	50	Electronics	Electronics
14-OCT-UK	14-OCT	14-10-2026	UK	Hiking Boots	Hiking Boots	Timberland	Hiking Boots Timberland	130	$130.00	10	Outdoor	Outdoor
17-JUN-IN	17-JUN	17-06-2026	IN	Laptop	Laptop	HP	Laptop HP	950	$950.00	25	Electronics	Electronics
25-NOV-AU	25-NOV	25-11-2026	AU	Sneakers	Sneakers	Adidas	Sneakers Adidas	90	$90.00	40		Unknown
08-DEC-DE	08-DEC	08-12-2026	DE	Coffee Maker	Coffee Maker	Nespresso	Coffee Maker Nespresso	120	$120.00	35		Unknown
18-FEB-CA	18-FEB	18-02-2026	CA	Smartwatch	Smartwatch	Fitbit	Smartwatch Fitbit	150	$150.00	15	Electronics	Electronics
16-APR-ES	16-APR	16-04-2026	ES	headphones	Headphones	Bose	Headphones Bose	250	$250.00	20	Electronics	Electronics
21-AUG-CA	21-AUG	21-08-2026	CA	Laptop Bag	Laptop Bag	Samsonite	Laptop Bag Samsonite	50	$50.00	35	Accessories	Accessories
20-AUG-CN	20-AUG	20-08-2026	CN	Smartwatch	Smartwatch	Huawei	Smartwatch Huawei	160	$160.00	15	Electronics	Electronics
27-JAN-IT	27-JAN	27-01-2026	IT	laptop	Laptop	Asus	Laptop Asus	980	$980.00	10	Electronics	Electronics
01-MAR-UK	01-MAR	01-03-2026	UK	Sunglasses	Sunglasses	Oakley	Sunglasses Oakley	150	$150.00	15	Fashion	Fashion
14-AUG-US	14-AUG	14-08-2026	US	Camping Tent	Camping Tent	Coleman	Camping Tent Coleman		$130.00	10	Outdoor	Outdoor
14-MAY-RU	14-MAY	14-05-2026	RU	Camera	Camera	Nikon	Camera Nikon	700	$700.00	50	Electronics	Electronics
09-JAN-CA	09-JAN	09-01-2026	CA	Microwave	Microwave	Panasonic	Microwave Panasonic	80	$80.00	20	Kitchen	Kitchen
19-JUL-BR	19-JUL	19-07-2026	BR	Fitness Tracker	Fitness Tracker	Xiaomi	Fitness Tracker Xiaomi	150	$150.00	30		Unknown
29-SEP-CA	29-SEP	29-09-2026	CA	Smartphone	Smartphone	Google	Smartphone Google	800	$800.00	45	Electronics	Electronics
03-JUN-CA	03-JUN	03-06-2026	CA	Sunglasses	Sunglasses	Ray-Ban	Sunglasses Ray-Ban		$130.00	25	Fashion	Fashion

6. Conditional Formatting:

● Apply data bar or color scales conditional formatting in the "Price" column.

● Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."



Product ID	Manufacturing date with year	Country code	Product Name Cleaning	Brand Name	Product Brand	Price ($) CLEANING	Quantity	Category CLEANING
28-JAN-US	28-01-2026	US	Laptop	Dell	Laptop Dell	$1,000.00	30	Electronics
15-FEB-US	15-02-2026	US	Sneakers	Nike	Sneakers Nike	$80.00	15	Fashion
03-MAR-US	03-03-2026	US	Coffee Maker	Keurig	Coffee Maker Keurig	$130.00	40	Kitchen
11-APR-US	11-04-2026	US	Smartphone	Samsung	Smartphone Samsung	$900.00	25	Electronics
22-MAY-US	22-05-2026	US	Backpack	North Face	Backpack North Face	$70.00	20	Unknown
07-JUN-UK	07-06-2026	UK	Headphones	Sony	Headphones Sony	$130.00	45	Electronics
19-JUL-UK	19-07-2026	UK	T-Shirt	Adidas	T-Shirt Adidas	$30.00	5	Fashion
23-AUG-UK	23-08-2026	UK	Blender	Ninja	Blender Ninja	$90.00	35	Kitchen
05-SEP-UK	05-09-2026	UK	Tablet	Apple	Tablet Apple	$500.00	50	Electronics
14-OCT-UK	14-10-2026	UK	Hiking Boots	Timberland	Hiking Boots Timberland	$130.00	10	Outdoor
17-JUN-IN	17-06-2026	IN	Laptop	HP	Laptop HP	$950.00	25	Electronics
25-NOV-AU	25-11-2026	AU	Sneakers	Adidas	Sneakers Adidas	$90.00	40	Unknown
08-DEC-DE	08-12-2026	DE	Coffee Maker	Nespresso	Coffee Maker Nespresso	$120.00	35	Unknown
18-FEB-CA	18-02-2026	CA	Smartwatch	Fitbit	Smartwatch Fitbit	$150.00	15	Electronics
16-APR-ES	16-04-2026	ES	Headphones	Bose	Headphones Bose	$250.00	20	Electronics
21-AUG-CA	21-08-2026	CA	Laptop Bag	Samsonite	Laptop Bag Samsonite	$50.00	35	Accessories
20-AUG-CN	20-08-2026	CN	Smartwatch	Huawei	Smartwatch Huawei	$160.00	15	Electronics
27-JAN-IT	27-01-2026	IT	Laptop	Asus	Laptop Asus	$980.00	10	Electronics
01-MAR-UK	01-03-2026	UK	Sunglasses	Oakley	Sunglasses Oakley	$150.00	15	Fashion
14-AUG-US	14-08-2026	US	Camping Tent	Coleman	Camping Tent Coleman	$130.00	10	Outdoor
14-MAY-RU	14-05-2026	RU	Camera	Nikon	Camera Nikon	$700.00	50	Electronics
09-JAN-CA	09-01-2026	CA	Microwave	Panasonic	Microwave Panasonic	$80.00	20	Kitchen
19-JUL-BR	19-07-2026	BR	Fitness Tracker	Xiaomi	Fitness Tracker Xiaomi	$150.00	30	Unknown
29-SEP-CA	29-09-2026	CA	Smartphone	Google	Smartphone Google	$800.00	45	Electronics
03-JUN-CA	03-06-2026	CA	Sunglasses	Ray-Ban	Sunglasses Ray-Ban	$130.00	25	Fashion